

A Cybernetic and Information-Theoretic Framework for the Transmission of Mythic Motifs in the Eastern Mediterranean (3000–700 BCE)

The transmission of religious and mythic narratives from Bronze Age Mesopotamia to Archaic Greece represents one of the most complex instances of asynchronous data transfer in human history. Traditionally, comparative mythology and classical archaeology have approached this phenomenon through the lens of linear diffusion, tracing the movement of deities and narratives as fixed artifacts migrating across the Mediterranean basin. This approach, while foundational, fails to account for the dynamic, highly degradative nature of ancient communication networks. A rigorous analysis of this historical process requires a fundamental paradigm shift: treating cultural transmission as a noisy, lossy communication channel operating within a complex, self-regulating cybernetic system.

By applying principles from Shannon-Hartley information theory, state-space topologies, quantum cognition, and machine learning, this report exhaustively models the cultural transmission network that operated between 3000 BCE and 700 BCE. The transmission pipeline—originating in the river valleys of Sumer and Akkad, bottlenecking in the Levantine coastal city-states and the island of Cyprus, and ultimately terminating in Mycenaean and Archaic Greece—did not merely pass data passively. Rather, it subjected the data to extreme environmental noise, homeostatic cybernetic feedback loops, and sequential observer effects that fundamentally altered the semantic structures of the myths themselves. This analysis defines the systemic parameters of this network, formalizes the narrative state transitions mathematically, and provides algorithmic mitigation strategies utilizing Bayesian inference and deep neural networks to strip away the distortions introduced by historical scribes, random archaeological survival, and modern teleological bias.

1. Informational Sets and System Architecture

To formalize the transmission of mythic motifs computationally, it is first necessary to define the informational sets that govern the system. These sets represent the foundational data structures manipulated, encoded, and decoded within the Eastern Mediterranean network. By defining these parameters, we establish the variables required for subsequent mathematical modeling. The primary data structure is the Mytheme Set, denoted as $S_{\{\text{myth}\}}$. This set comprises the discrete, fundamental units of narrative and theological data. Examples of elements within $S_{\{\text{myth}\}}$ include the *Chaoskampf* (the motif of a storm god defeating a primordial sea serpent, such as Tarhunt and Illuyanka, or Zeus and Typhon), the castration of the primordial sky father, or the descent of a fertility and war goddess into the underworld. These motifs are not static; they are highly malleable vectors of cultural information. The second required structure is the Linguistic Protocol Set, denoted as $S_{\{\text{lang}\}}$. This set encompasses the encoding mechanisms used to package $S_{\{\text{myth}\}}$ for transmission across the network. The protocols include the logo-syllabic cuneiform scripts of Sumerian, Akkadian, Hurrian, and Hittite, the alphabetic scripts

of Phoenician, and the syllabic and alphabetic systems of early Greek, including Mycenaean Linear B and the later Euboean alphabet.

The final, and perhaps most critical, data structure is the Observer Matrix, denoted as $S_{\{obs\}}$. This matrix represents the localized cognitive biases, political exigencies, theological frameworks, and sensory limitations of the entities receiving, decoding, and re-encoding the data at each node. $S_{\{obs\}}$ acts as an active filter, permanently altering the data payload of $S_{\{myth\}}$ as it transitions between different elements of $S_{\{lang\}}$.

1.1 The Narrative Block Diagram and Network Topology

The geographic and cultural topography of the Eastern Mediterranean acts as the physical hardware of this communication channel. The system architecture can be visualized as a Directed Acyclic Graph (DAG) with intermittent cyclic feedback loops, where nodes process data, edges transmit it, and filters degrade or alter it. The structural components of this network are delineated below.

System Component	Elements and Variables	Function within the Communication Channel
Nodes (Transceivers)	$N_{\{Mesopotamia\}}$ (Sumer, Akkad), $N_{\{Anatolia\}}$ (Hatti, Wilusa, Millawanda), $N_{\{Levant\}}$ (Ugarit, Sidon, Tyre, Byblos), $N_{\{Cyprus\}}$ (Kition, Paphos, Amathous), $N_{\{Greece\}}$ (Mycenae, Euboea, Athens, Sparta).	Serve as localized data processing centers where incoming mythic signals are received, decoded, stored in local memory (oral or written), modified via cybernetic feedback to fit local homeostasis, and re-transmitted.
Edges (Channels)	Maritime Trade Routes (Copper, Tin, Incense), Diplomatic Exchange (e.g., Tawagalawa Letter, Amarna Letters), Warfare, and Migration (e.g., the Sea Peoples).	The physical medium of data transfer. Channel bandwidth is directly proportional to the volume of trade, the frequency of diplomatic interaction, and the proximity of the interacting nodes.
Filters (Noise)	Linguistic Translation, Morphological Drift, Geographic Isolation, and Catastrophic Systemic Collapse.	Introduces entropy into the signal. Data loses high-fidelity theological nuances when translated across entirely different language families (e.g., from Semitic Akkadian to Indo-European Greek) or when the physical medium of transmission degrades.

1.2 Fixed Boundaries and the 1185 BCE Anchor Point

In computational modeling of historical systems, fixed chronological anchor points are required to bound the algorithmic timeline and define state transitions. The destruction of the Levantine port city of Ugarit in approximately 1185 BCE serves as a critical phase transition in the network

topology. Prior to this boundary, the Late Bronze Age Mediterranean functioned as a highly interconnected, high-bandwidth network characterized by standardized diplomatic protocols, extensive maritime trade, and massive palatial nodes utilizing Akkadian as a diplomatic lingua franca. In this environment, complex theological narratives could be transmitted with relative stability between elite centers.

Following the Bronze Age Collapse, triggered by the migrations of the Sea Peoples, systemic droughts, and the fall of the Hittite Empire, the network topology shattered. The destruction of primary routing nodes like Ugarit and Hattusa forced the communication channel into a highly decentralized, low-bandwidth state. During the subsequent Greek Dark Ages, the primary medium of data storage in the Aegean reverted from rigid, palace-controlled syllabic scripts (Linear B) back to fluid oral traditions. This severe fluctuation in channel capacity fundamentally altered how motifs survived. For instance, the Hurrian *Kumarbi* cycle, which details a violent succession of gods, was preserved only in structural memory, eventually reaching Hesiod in Archaic Greece heavily degraded of its original Anatolian context. The 1185 BCE anchor point thus represents the transition from a structured digital-equivalent network to an analog, highly noisy distributed network.

2. Theoretical Framework and Mathematical Modeling

The transmission of a complex myth across centuries, oceans, and language barriers is perfectly analogous to the transmission of a signal across a degrading communication channel. By mapping these cultural phenomena to formal mathematical frameworks, the evolution of religious narratives transforms from a subjective literary and philological study into a rigorous computational science.

2.1 Information Entropy and Channel Capacity (Shannon-Hartley)

The transmission of a myth from one culture to another can be quantified using the foundational principles of Claude Shannon's Information Theory. Let a mythic narrative X possess an inherent information entropy $H(X)$, representing the total complexity of its motifs, theological paradoxes, ritual requirements, and socio-political context. The capacity C of the transmission channel—such as Phoenician merchants interacting with Greek elites at Cyprus—is determined by the Shannon-Hartley theorem:

$$C = B \log_2 \left(1 + \frac{S}{N} \right)$$

In this cultural application, B represents the bandwidth, defined as the frequency, duration, and depth of cultural contact. S is the signal power, representing the cultural dominance, economic leverage, or religious zeal of the transmitting node. N represents the ambient noise, which includes linguistic barriers, temporal distance, and the physical loss of texts or artifacts.

In the specific transmission of the Mesopotamian goddess Inanna/Ishtar through the Levant to become Astarte, and subsequently arriving in Cyprus to become Aphrodite, the channel capacity was heavily restricted by geographical and cultural bottlenecks. The island of Cyprus (N_{Cyprus}) acted as a crucial repeater station and processing node. At major Cypriot centers like Paphos, Amathous, and Kition, Phoenician merchants established sanctuaries dedicated to Astarte. Here, the Phoenician signal merged with a native Cypriot fertility deity, creating a highly syncretic data packet known to the Greeks as *Aphrodite Paphia*.

Because the inherent complexity of the original Mesopotamian narrative was greater than the capacity of the Aegean channel ($H(X) > C$), the data transfer was fundamentally lossy. The

principles of Rate-Distortion Theory dictate that when the required transmission rate exceeds channel capacity, the signal must undergo lossy compression, minimizing the bit rate while keeping distortion within an acceptable threshold. The rate-distortion function $R(D)$ defines this minimum transmission rate needed to ensure that the distortion does not exceed a localized tolerance D :

$$R(D) = \min_{p(y|x): \sum p(x,y)d(x,y) \leq D} I(X;Y)$$

When Phoenician traders compressed the highly complex, paradoxical theology of Ishtar and Astarte—who embodied both fierce, bloodthirsty warfare and erotic love—into a signal capable of crossing the Aegean, the Greek receiving nodes minimized the distortion D by filtering the data through their own pre-existing cultural templates. Consequently, much of Astarte's martial, aggressive nature was discarded in the standard Hellenic pantheon. The Greek homeostatic system routed the martial portfolio primarily to Ares and Athena, while Aphrodite retained the erotic, maritime, and celestial (Venus) parameters. Stephanie Budin's extensive critique of the over-simplification of Aphrodite's origins highlights exactly this phenomenon of lossy compression: modern observers often fail to recognize that the transmission was not a perfect one-to-one copy, but a severely distorted reconstruction tailored to the limited cultural bandwidth of Archaic Greece.

2.2 State-Space Transitions and Vector Mathematics

To map narrative drift systematically, core narrative motifs must be defined as vectors in a high-dimensional state space V . Let a specific deity or myth m at time t be represented as a state vector $\vec{m}_t$ in a vector space where each orthogonal dimension corresponds to a distinct cultural or theological attribute. For example, a deity vector could be defined by dimensions corresponding to x_1 = agricultural fertility, x_2 = maritime protection, x_3 = celestial/astral association, and x_4 = martial prowess.

The transmission of a myth from Culture A to Culture B is modeled by an algorithmic transition matrix $T_{\{A \rightarrow B\}}$. The resulting syncretic myth in Culture B is mathematically defined as:

$$\vec{m}_{t+1} = T_{\{A \rightarrow B\}} \vec{m}_t + \vec{\epsilon}$$

Where $\vec{\epsilon}$ is an error vector introduced by transmission noise.

Consider the Hurrian *Song of Kumarbi*, a defining mythological text of the Late Bronze Age. The narrative details the primordial sky god Anu being violently overthrown and castrated by Kumarbi, who subsequently gives birth to the storm god Teshub from the swallowed genitals. This text was transmitted to the Hittites, encoded in their cuneiform archives at Hattusa, and eventually, centuries later, reached Archaic Greece, where it was structurally reconstructed in Hesiod's *Theogony* as the castration of Uranus by Kronos, leading to the rise of Zeus. The transition matrix $T_{\{\text{Hurrian} \rightarrow \text{Greek}\}}$ heavily weighted and prioritized the intergenerational succession and castration dimensions of the state vector. Conversely, the dimensions corresponding to specific Anatolian geographic anchors and minor Hurrian attendant deities decayed to zero, stripped away by the error vector $\vec{\epsilon}$ as the myth traversed the post-collapse Aegean.

2.3 Cybernetic Feedback Loops and Cultural Homeostasis

Absorbing cultures do not act as passive receivers of information; they operate as complex, self-regulating cybernetic systems. According to the foundational principles of cybernetics, biological and social systems adjust their internal environments to maintain a stable state, or homeostasis, despite constant external perturbations.

When a foreign mythic motif enters a localized socio-political stability—such as the highly centralized, rigid Mycenaean palatial economy—it acts as an external perturbation. To prevent semantic or theological destabilization, the absorbing node employs a homeostatic feedback loop to actively alter the myth's trajectory, neutralizing elements that conflict with local power structures.

The Tawagalawa letter (AhT 4) provides explicit historical documentation of this phenomenon. The text records direct diplomatic and military interactions between the Hittite Great King (likely Hattusili III) and the King of Ahhiyawa (Mycenaean Greece) over western Anatolian territories, specifically Wilusa (Troy) and Millawanda (Miletus). The letter illustrates a sophisticated diplomatic channel where the Hittite King addresses the Ahhiyawan ruler as "My Brother" and a "Great King"—though the latter title was notably erased in the surviving draft, indicating active calibration of diplomatic status. In this high-stakes environment, the exchange of cultural and mythic data was fiercely regulated. The Hittites systematically incorporated foreign gods into their pantheon, known as the "Thousand Gods of Hatti," as a deliberate administrative mechanism to pacify conquered or adjacent territories. By mapping a foreign deity onto a local one—for instance, recognizing a Mycenaean or Luwian equivalent of a storm god and absorbing their cultic practices—the Hittite state neutralized the foreign theological threat, utilizing a negative feedback loop to maintain supreme state equilibrium.

3. The Observer Effect and Signal Distortion

In quantum mechanics, the act of observation fundamentally and irreversibly alters the state of the system being observed. This principle is strictly applicable to computational history. The data matrix representing ancient myth is continuously distorted by the very mechanisms utilized to record, excavate, and interpret it. This "Observer Effect" manifests sequentially in three distinct temporal phases, permanently biasing the informational pipeline.

3.1 The Historical Observer: Quantum Cognition and State Collapse

Prior to being codified in writing, oral traditions exist in a state of high entropy and narrative fluidity. In the emerging framework of Quantum Cognition, mental states and highly ambiguous semantic structures do not adhere to classical deterministic logic; rather, they are modeled using the probability calculus of quantum mechanics, where human cognition can hold contradictory beliefs or interpretations simultaneously. An oral myth, transmitted by wandering bards over centuries, exists in a state of "Semantic Superposition". It contains multiple, regional, and sometimes contradictory variants simultaneously. For example, Aphrodite exists in superposition as both the ancient daughter of Zeus and Dione (as in Homer) and an ancient primordial entity born of sea-foam and Uranus's severed genitals (as in Hesiod).

The historical scribe acts as the measurement apparatus. The transition from a fluid, dynamic oral tradition to a rigid, baked cuneiform tablet or alphabetic papyrus acts precisely as a projective measurement in quantum theory. In quantum decision theory, this is formalized using a Hilbert space $\mathcal{H}$ where the unwritten myth is a state vector $|\psi\rangle$ representing a superposition of all possible narrative variants. When a scribe in Ugarit, Hattusa, or Athens decides to commit the myth to writing, they apply an observable operator M . The physical act of writing "collapses" the probability wave into a single, definitive eigenstate, mathematically eliminating all other regional variations from the immediate historical record:

$$|\psi\rangle \xrightarrow{\text{Observation}} |\phi_k\rangle$$

Crucially, this collapse is heavily biased by the elite priorities of the scribe. The historical observer systematically filters out narratives favored by the peasantry, rival cults, or disenfranchised groups, permanently biasing the informational set $S_{\{\text{myth}\}}$ toward palatial administration and theocratic legitimization.

3.2 The Archaeological Observer: Survival Bias and Matrix Completion

Once collapsed into a physical medium, the signal is subjected to the extreme stochastic noise of archaeological survival over millennia. This introduces the second observer effect: survival bias. The laws of material degradation act as a highly asymmetric environmental filter. Clay tablets in Mesopotamia and Anatolia, ironically preserved by the catastrophic fires that destroyed their cities, survive at a much higher rate than the papyrus, parchment, or wooden boards utilized extensively in the Levant, Egypt, and early Greece.

Mathematically, the surviving historical record is not a complete dataset; it is a severely incomplete observation matrix O . The true historical matrix X of size $n \times p$ (where n is the number of actual historical events and p is the number of cultural attributes) is subjected to a masking operator M . In this binary mask, $M_{\{ij\}} = 1$ if the artifact survives to be excavated, and 0 if it is destroyed by time or human activity. The data available to the modern epigrapher or archaeologist is the Hadamard product:

$$O = X \circ M$$

Because the masking matrix M is non-randomly distributed—highly skewed toward inorganic materials and hyper-arid climates—the matrix O contains immense structural gaps. When modern scholars attempt to reconstruct the true historical matrix X from O using cognitive or computational matrix completion algorithms, the missing data creates profound false correlations. If an epigrapher finds a specific motif in Hittite cuneiform texts and later in Greek alphabetic texts, but the intervening Phoenician papyri have rotted away completely, the missing data might falsely suggest a direct, unmediated cultural transfer, completely ignoring the crucial Levantine and Cypriot intermediary nodes.

3.3 The Modern Observer: Teleological Synthesis and Eurocentric Bias

The third observer effect occurs in the modern era, where scholars from the 19th to the 21st centuries impose synthetic, often teleological order upon a fundamentally chaotic and fragmented dataset. This bias effectively projects a linear "evolutionary" tree onto what was historically a highly reticulated, multidirectional network.

Early Eurocentric classical scholarship frequently attempted to partition "Eastern" and "Western" civilizations into discrete silos, downplaying the heavy Asiatic and Semitic influence on Greek culture. It was not until the synthesis of the "Orientalizing Revolution" by scholars like Walter Burkert and M.L. West that the academic consensus fully acknowledged the massive influx of Near Eastern motifs into the Aegean during the Early Archaic age.

Even with this recognition, modern observers tend to force complex syncretism into clean, linear narratives. This is evident in early attempts to equate the Hittite term Ahhiyawa directly and exclusively with a unified, Homeric Achaeans empire, ignoring the fragmented reality of Mycenaean palace-states. Similarly, assuming Aphrodite is merely a direct Greek copy of the Phoenician Astarte severely oversimplifies the data. As Stephanie Budin meticulously notes in

her critique of the "sacred prostitution" myth, early classical scholars hallucinated an entire, pervasive institutional practice in the ancient Near East by misinterpreting fragments of Herodotus and Strabo through a highly biased Greco-Roman lens. The modern scholarly framework acts as a final filter, F_{modern} , which minimizes perceived entropy by artificially smoothing out contradictions. It forces a complex Directed Acyclic Graph with numerous lateral transfers into a simple, bifurcating phylogenetic tree, synthesizing order out of noise at the cost of historical truth.

4. Mathematical Formalizations for Network Topology

To rigorously track the transmission of a mytheme and strip away the compounding noise of the observer effects, computational historians must deploy advanced algorithmic mitigation strategies. Simple linear timelines and comparative philology are insufficient; the dataset must be processed using the mathematics of network theory and computational phylogeny.

4.1 Markov Chains and Directed Acyclic Graphs

The trajectory of a mythic motif through the Eastern Mediterranean cannot be modeled as a memoryless system. A first-order Markov chain assumes that the probability of transitioning to the next state depends only on the current state. However, cultural transmission retains deep historical memory. Therefore, the system must be modeled using a higher-order generalization. A second-order Markov chain is utilized to model the sequence of transmission, where the probability of a motif transitioning to a specific state in Archaic Greece (x_{t+1}) depends not only on its immediate prior state in Cyprus (x_t), but also on its foundational state in the Levant (x_{t-1}). This is defined as:

$$P(X_{t+1} = x \mid X_t = y, X_{t-1} = z)$$

This formalization captures the reality that Greek syncretism was influenced by both immediate Phoenician contact and older, deeply embedded Bronze Age memories.

4.2 Computational Phylogeny and NeighborNet Algorithms

Tracking the transmission of a mytheme requires abandoning simple bifurcating trees in favor of computational phylogenetic networks. In evolutionary biology, phylogenetic trees assume a new trait appears once and diverges linearly. However, cultural narratives experience massive horizontal transmission—myths borrow elements from neighboring cultures long after their initial divergence, requiring graphs that support cycles and lateral edges.

Algorithms such as NeighborNet generate split graphs that visualize these complex, non-treelike evolutionary histories. By encoding mythemes as binary characters (present or absent) across different mythological variants, researchers generate a proximity matrix. When applied to ancient structural motifs—such as the "Cosmic Hunt" or the "Struggle for Supremacy"—phylogenetic networks demonstrate that global mythologies are hierarchically nested and correlate heavily with human migration patterns out of specific geographic bottlenecks, flowing from the Near East into Eurasia and the Americas.

5. Observer Mitigation Strategies and Machine

Learning

To isolate the raw historical signal from the noise introduced by the scribe, archaeological decay, and the modern scholar, computational history relies heavily on artificial intelligence, machine learning embeddings, and Bayesian inference.

5.1 Machine Learning Embeddings and Semantic Drift

Currently, computational linguists are utilizing deep neural networks to process massive datasets of fragmented ancient texts. The Cuneiform Digital Library Initiative (CDLI), hosting over 360,000 digitized and transliterated clay tablets, serves as the primary training ground for Convolutional Neural Networks (CNNs) and transformer-based architectures.

These machine learning algorithms utilize high-dimensional word embeddings—such as word2vec or BERT architectures modified for Akkadian and Sumerian—to map semantic shifts. In a vector embedding space, words with similar contextual meanings cluster together. By calculating the cosine similarity between the vector representation of a word in Akkadian and its translated equivalent in Linear B or Archaic Greek, researchers can quantitatively measure the semantic drift of a theological concept across centuries.

For example, tracking the epithets of the Venus goddess (Ishtar vs. Aphrodite) through an embedding space reveals exactly which attributes were preserved and which were discarded during the translation across the Aegean. The algorithms detect continuous transformations in the state-space that human philologists, reliant on discrete and static vocabulary lists, might entirely overlook. Furthermore, these NLP models are achieving breakthrough success in direct Cuneiform-to-English translation, scoring highly on the BLEU4 evaluation metric, allowing for rapid, unbiased processing of raw logographic data.

5.2 Error-Correcting Codes and Deep Neural Networks

Advanced AI tools currently act as error-correcting codes to address the missing data in the observation matrix O . DeepMind's *Ithaca*, published in *Nature*, represents a watershed moment in this field. *Ithaca* is a deep neural network specifically designed for the textual restoration, geographical attribution, and chronological attribution of ancient Greek inscriptions.

By processing the input text as joint character and word representations, and representing damaged or missing characters with a special <unk> token, *Ithaca* contextualizes damaged strings of text. The transformer architecture then hallucinates the missing data with distinct statistical probabilities. While the AI alone achieves 62% accuracy in restoring damaged texts, human historians using *Ithaca* see their accuracy leap from 25% to 72%. DeepMind's subsequent iteration, *Aeneas*, further advances this by retrieving textual and contextual parallels, predicting missing text even when the gap length is entirely unknown. By deploying models like *Ithaca* and *Aeneas* on epigraphic datasets, computational historians can algorithmically fill the gaps caused by survival bias, restoring a higher-fidelity picture of the raw historical signal.

5.3 Bayesian Inference for Bias Mitigation

To strip away the modern and historical observer biases mathematically, Bayesian inference offers a rigorous framework. Bayesian models update the probability of a historical hypothesis

(e.g., that the Greek castration myth originated specifically in Hurrian theology) as more evidence becomes available, explicitly accounting for the reliability of the evidence.

Let H be the historical truth (the unobserved state of the myth) and O be the observed archaeological text. The posterior probability of the historical truth given the text is:

$$P(H \mid O) = \frac{P(O \mid H) P(H)}{P(O)}$$

The term $P(O \mid H)$ serves as the error model, explicitly defining the probability of an observer bias or survival bias distorting the text. If the observation O is a pristine marble inscription from Classical Athens—a medium highly susceptible to state-sponsored propaganda—the likelihood $P(O \mid H)$ is dynamically adjusted to reflect a high degree of political noise. Conversely, an accounting ledger on a clay tablet may have a lower noise coefficient for ideological bias.

6. Algorithmic Reconstruction: Two Case Studies

To demonstrate the efficacy of this cybernetic and information-theoretic framework, we apply it to two major mythic transmission pipelines crossing the Late Bronze Age Mediterranean.

6.1 The Venus-Goddess Trajectory: Multi-Node Filtering

The transmission of the Great Goddess motif from Mesopotamia to Greece is a premier example of multi-hop network routing characterized by heavy signal degradation and localized homeostatic syncretism.

1. **Node $N_{\{\text{Mesopotamia}\}}$ (Source):** Inanna/Ishtar originates as a highly complex vector containing parameters for visceral warfare, erotic love, celestial governance (the planet Venus), and political legitimization of the Akkadian kings.
2. **Node $N_{\{\text{Levant}\}}$ (Routing & Modification):** Transmitted to the coastal mercantile hubs of Ugarit, Sidon, and Tyre, the vector shifts to Astarte (Ashtart). The goddess retains the martial and sexual parameters, often depicted iconographically with lions and doves, but gains a profound association with maritime navigation—a mandatory homeostatic adaptation for the seafaring Phoenicians.
3. **Node $N_{\{\text{Cyprus}\}}$ (Syncretic Bottleneck):** Astarte is transmitted via Phoenician trade routes to major Cypriot port cities. The archaeological record at the Bamboula sanctuary in Kition and the grand temple at Paphos shows an intense synthesis. The $S_{\{\text{obs}\}}$ of Cyprus filters the signal, establishing large sanctuary complexes where the Phoenician deity merges with an indigenous fertility goddess, worshipped simply as *Anassa* (the Sovereign) or the *Paphian*.
4. **Node $N_{\{\text{Greece}\}}$ (Terminal Receiver):** The Cypriot syncretic packet crosses the Aegean to Cythera and the mainland. By the time it is decoded by the Greeks, the vector has undergone severe lossy compression. The Hellenic homeostatic feedback loop rejects the synthesis of supreme martial power and erotic love within a single female deity. The martial parameters are ruthlessly filtered out. Aphrodite emerges as *Ourania* (heavenly) and *Pandemos* (of the people), retaining the doves, the maritime associations, and the erotic portfolio, but completely stripped of Ishtar's bloodlust.

6.2 The Succession Myth: High-Fidelity Diplomatic Transfer

In stark contrast to the slow, multi-hop diffusion and degradation of the Venus goddess, the

transmission of the Succession Myth demonstrates a much more direct, high-fidelity transfer mechanism utilizing high-bandwidth diplomatic channels.

1. **Source Signal:** The Hurrian *Song of Kumarbi* details a violent theogony: the primordial sky god Anu is overthrown by Kumarbi, who bites off Anu's genitals, subsequently giving birth to the storm god Teshub.
2. **Channel:** The Hittite Empire, acting as a massive cultural sponge, absorbs Hurrian theology to stabilize its southern and eastern provinces. The myth is translated into Hittite cuneiform, preserving the high-entropy details due to the formal, rigid nature of palatial scribal schools.
3. **Transmission Edge:** The intense diplomatic and military contact between the Hittites and the Ahhiyawa (Mycenaeans) in western Anatolia, as evidenced by texts like the Tawagalawa letter, provides a high-bandwidth channel for ideological and cultural exchange.
4. **Receiving Node:** The Mycenaeans absorb the structural narrative. Despite the catastrophic systemic collapse of 1185 BCE, the core topological structure of the myth (Sky Father → Castrator → Storm God) proves highly resilient to memory degradation. It survives the oral traditions of the Greek Dark Ages to be re-collapsed into a text by Hesiod in the 8th century BCE. The sequence perfectly mirrors the original: Uranus → Kronos → Zeus. The structural integrity of this myth proves that high-level state contacts can transmit complex informational topologies with minimal distortion, even across the phase transition of systemic societal collapse.

Works cited

1. Full text of "Greek Mythology 1 - Deities" - Internet Archive, https://archive.org/stream/GreekMythology1-Deities/GreekMythology-1_djvu.txt
2. Kumarbi - Wikipedia, <https://en.wikipedia.org/wiki/Kumarbi>
3. Hittite Myths 2nd Edition Writings From The Ancient World HKQJHT Elifile - Ir - Scribd, <https://www.scribd.com/document/996784163/Hittite-Myths-2nd-Edition-Writings-From-the-Ancient-World-Hkqjht-Elifile-ir>
4. The transmission of the alphabet to the Aegean, in Ł. Niesiołowski-Spanò and M. Węcowski (eds), *Change, Continuity and Connectivity. North-Eastern Mediterranean at the turn of the Bronze Age and in the early Iron Age*, Wiesbaden 2018: Harrassowitz Verlag, 235–257 - Academia.edu, https://www.academia.edu/37781464/The_transmission_of_the_alphabet_to_the_Aegean_in_%C5%81_Niesio%C5%82owski_Span%C3%B2_and_M_W%C4%99cowski_eds_Change_Continuity_and_Connectivity_North_Eastern_Mediterranean_at_the_turn_of_the_Bronze_Age_and_in_the_early_Iron_Age_Wiesbaden_2018_Harrassowitz_Verlag_235_257
5. 1 Introduction - arXiv, <https://arxiv.org/html/2406.04039v1>
6. Translating Akkadian to English with neural machine translation | PNAS Nexus, <https://academic.oup.com/pnasnexus/article/2/5/pgad096/7147349>
7. The Ahhiyawa Question: Reconsidered | Ağustos 2021, Cilt 85 - Sayı 303 - Belleten, <https://belleten.gov.tr/tam-metin/3654/eng>
8. What Did the Hittites Write About the Trojan War? - TheCollector, <https://www.thecollector.com/hittites-trojan-war/>
9. Egyptian Parallels for an Incident in Hesiod's Theogony and an Episode in the Kumarbi Myth, https://www.academia.edu/9967593/Egyptian_Parallels_for_an_Incident_in_Hesiods_Theogony_and_an_Episode_in_the_Kumarbi_Myth
10. Teshub - Religion Wiki - Fandom, <https://religion.fandom.com/wiki/Teshub>
11. Knowledge Decay. Parallels Between Deep Learning and... | by Daniel Rodríguez | sadasant | Medium,

<https://medium.com/sadasant/knowledge-decay-024b6c2c2109> 12. Gods and Goddesses of Ancient Egypt | PDF - Scribd,
<https://www.scribd.com/document/320040022/154978105-Gods-and-Goddesses-of-Ancient-Egypt> 13. Anat - Wikipedia, <https://en.wikipedia.org/wiki/Anat> 14. CYPRUS: Myth and Cult of Aphrodite on Cyprus - earthstoriez,
<https://earthstoriez.com/cyprus-myth-and-cult-of-aphrodite-on-cyprus> 15. Fia550.research paper | PPTX - Slideshare, <https://www.slideshare.net/slideshow/fia550research-paper/3951340> 16. Astarte - Wikipedia, <https://en.wikipedia.org/wiki/Astarte> 17. Rate-Distortion Theory - ResearchGate, https://www.researchgate.net/publication/229817419_Rate-Distortion_Theory 18. Introduction to Data Compression | MBIT,
https://www.mbit.edu.in/wp-content/uploads/2020/05/data_compression.pdf 19. A Reconsideration of the Aphrodite-Ashtart Syncretism - ResearchGate,
https://www.researchgate.net/publication/233544763_A_Reconsideration_of_the_Aphrodite-Ashtart_Syncretism 20. Brill's Companion to Aphrodite - Bryn Mawr Classical Review,
<https://bmcr.brynmawr.edu/2012/2012.07.13/> 21. The Myth of Sacred Prostitution in Antiquity - Cambridge University Press & Assessment,
<https://www.cambridge.org/core/books/myth-of-sacred-prostitution-in-antiquity/61198555C0E969D82AC0B8900F7D99E7> 22. The Myth of Sacred Prostitution in Antiquity - Bryn Mawr Classical Review, <https://bmcr.brynmawr.edu/2009/2009.04.28/> 23. Karl FRISTON | University College London, London | UCL | Institute of Neurology | Research profile - ResearchGate,
<https://www.researchgate.net/profile/Karl-Friston> 24. estratto - Aegeus Society,
<https://www.aegeussociety.org/wp-content/uploads/2021/11/Bryce-2018-Ahhiyawa.pdf> 25. The Trojan War #3: Bronze Age evidence - Kiwi Hellenist,
<https://kiwihellenist.blogspot.com/2016/09/the-trojan-war-3-bronze-age-evidence.html> 26. The Ahhiyawa World Behind Homer's Odyssey | by Bill Giannakopoulos - Medium,
<https://medium.com/@bill.giannakopoulos/the-ahhiyawa-world-behind-homers-odyssey-e96f54240d80> 27. THE CHARIOTS OF AHHIYAWA1 - DACIA,
https://www.daciajournal.ro/pdf/DACIA%202004_2005/art12Kelder.pdf 28. ANE TODAY – 201911 – A Great King and a Wanax? The Politics of Mycenaean Greece,
<https://www.asor.org/anetoday/2019/11/Great-King-and-a-Wanax> 29. An overview of the quantum cognition research programme - City Research Online,
<https://openaccess.city.ac.uk/id/eprint/34770/> 30. Cognition in Superposition: Quantum Models in AI, Finance, Defence, Gaming and Collective Behaviour - arXiv,
<https://arxiv.org/html/2508.20098v1> 31. A quantum-cognitive approach to dynamic meaning construction - Frontiers,
<https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2026.1664747/full> 32. Quantum Models of Cognition and Decision | Request PDF - ResearchGate,
https://www.researchgate.net/publication/285942923_Quantum_Models_of_Cognition_and_Decision 33. Shaping History: Advanced Machine Learning Techniques for the Analysis and Dating of Cuneiform Tablets over Three Millennia - ResearchGate,
https://www.researchgate.net/publication/381226742_Shaping_History_Advanced_Machine_Learning_Techniques_for_the_Analysis_and_Dating_of_Cuneiform_Tablets_over_Three_Millennia 34. An overview of methods to deal with missing values - Julie Josse,
http://juliejosse.com/wp-content/uploads/2022/03/diableret_lecture.pdf 35. The East Face of Helicon: West Asiatic Elements in Greek Poetry and Myth (Clarendon Paperbacks) 0198152213, 9780198152217 - DOKUMEN.PUB,
<https://dokumen.pub/the-east-face-of-helicon-west-asiatic-elements-in-greek-poetry-and-myth-clarendon-paperbacks-0198152213-9780198152217.html> 36. A Companion to Ancient Education

- Cristo Raul.org,
<https://www.cristoraul.org/ENGLISH/readinghall/THIRDMILLENNIUMLIBRARY/PDF/A-Companion-to-Ancient-Education.pdf> 37. Wisdom Between East and West: Mesopotamia, Greece and Beyond,
<https://iris.unito.it/retrieve/e1ca94f9-2f45-4dae-88f8-f4ca986d22dc/Viano%20%26%20Sironi%202024%20-%20Wisdom%20between%20East%20and%20West.pdf> 38. Ahhiyawa, Hatti, and Diplomacy: Implications of Hittite Misperceptions of the Mycenaean World - ResearchGate,
https://www.researchgate.net/publication/352049959_Ahhiyawa_Hatti_and_Diplomacy_Implications_of_Hittite_Misperceptions_of_the_Mycenaean_World 39. Program for Wednesday, July 12th - EasyChair, <https://easychair.org/smart-program/NetSci2023/2023-07-12.html> 40. Evolving View on Phylogenetic Networks | Systematic Biology - Oxford Academic,
<https://academic.oup.com/sysbio/advance-article/doi/10.1093/sysbio/syag045/8729353> 41. A LARGE-SCALE STUDY OF WORLD MYTHS; pp. 407–424 - ResearchGate,
https://www.researchgate.net/publication/329098364_A_LARGE-SCALE_STUDY_OF_WORLD_MYTHS_pp_407-424 42. A split graph showing the results of NeighborNet analyses of the Cosmic Hunt versions. - ResearchGate,
https://www.researchgate.net/figure/A-split-graph-showing-the-results-of-NeighborNet-analyses-of-the-Cosmic-Hunt-versions_fig7_259193627 43. Deeply nested structure of mythological traditions worldwide - arXiv, <https://arxiv.org/html/2408.07300v1> 44. Evolution of Myths and Migration | PDF - Scribd, <https://www.scribd.com/document/900075961/Evolution-of-Myths> 45. Machine Learning for Ancient Languages: A Survey | Computational Linguistics | MIT Press,
<https://direct.mit.edu/coli/article/49/3/703/116160/Machine-Learning-for-Ancient-Languages-A-Survey> 46. Contributions to Computational Assyriology - Helda - University of Helsinki,
<https://helda.helsinki.fi/bitstreams/3b85f1ba-15ec-44a6-afcd-2db5ccd522a6/download> 47. Restoring and attributing ancient texts using deep neural networks - PubMed,
<https://pubmed.ncbi.nlm.nih.gov/35264762/> 48. (PDF) Restoring and attributing ancient texts using deep neural networks - ResearchGate,
https://www.researchgate.net/publication/359121987_Restoring_and_attributing_ancient_texts_using_deep_neural_networks 49. Predicting the past with Ithaca - Google DeepMind,
<https://deepmind.google/blog/predicting-the-past-with-ithaca/> 50. google-deepmind/ithaca: Restoring and attributing ancient texts using deep neural networks,
<https://github.com/google-deepmind/ithaca> 51. Research - Thea Sommerschield,
<https://theasommerschield.it/research.html> 52. Aeneas transforms how historians connect the past - Google DeepMind,
<https://deepmind.google/blog/aeneas-transforms-how-historians-connect-the-past/> 53. THE PENNSYLVANIA STATE UNIVERSITY SCHREYER HONORS COLLEGE GODDESSES OF THE SACRED HARBOR: MARITIME CULT, NAVIGATION, AND IDENTIT,
https://honors.libraries.psu.edu/files/final_submissions/10726?remediate_token=eyJfcMfPbHMiOnsibWVzc2FnZSI6IkJBaHBBdVlwiwiZXhwljoiMjAyNi0wNC0yM1QwMzowOTozOC42NzdaliwiChVyljoicmVtZWRpYXRlX3JlcXVlc3QifX0%3D--15c318b6e075ed625ae451fc64bd7234f92d47b2&remediated=false 54. Moving to a new place: the cult of Paphia Aphrodite in Hellenistic Nea Paphos,
https://www.academia.edu/37358040/Moving_to_a_new_place_the_cult_of_Paphia_Aphrodite_in_Hellenistic_Nea_Paphos